

Henry Lefebvre

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EXPERIENCE

- Research Assistant* Jan. 2025 - Sep. 2025
McGill University - Computer Graphics Lab
- Researched and implemented acceleration structures for real-time **physics-based animation** and collision detection on soft 3D models using **Python**
 - Designed a **novel algorithm** to update bounded deformation trees, improving bounding tightness by up to 50% over existing methods
 - Analyzed a dataset of **10,000 3D models** to evaluate algorithm performance
- Software Developer - Autonomous Underwater Vehicle* Dec. 2023 - June 2024
McGill University - Robotics
- Developed and tested **computer vision** solutions for a robotic submarine
 - Transformed live underwater camera feed into a navigable map of labeled objects in global coordinate space using object-detection algorithms
- Software Developer - MAIS Kernel* Dec. 2023 - June 2024
MAIS Kernel - McGill AI Society
- Used **AWS** and **Docker** to build an AI document analysis web app in **Django** to help local real-estate agencies manage rental agreements
- Data Science & Machine Learning Intern* March 2023 - Sep. 2023
M2M Tech
- Performed exploratory data analysis on financial datasets and created interactive visualizations to better understand market trends

EDUCATION

- McGill University** - B.Sc., Honours Computer Science Sep. 2022 - May 2026
GPA: 3.9 / 4.0
Coursework:
- Data Structures & Algorithms, Operating Systems, Software Design, Distributed Systems, Functional Programming, Numerical Computing, Machine Learning, Reinforcement Learning, Graphics, Databases, Compilers, Calculus, Probability, Statistics, Linear Algebra, Differential Equations

PROJECTS

- rt - Ray Tracer** June 2024 - Sep. 2024
- Developed a multithreaded **ray tracer** in Rust to render realistic 3D scenes containing meshes with high-resolution textures and millions of polygons
- ApartMatch** - **1st Place Hack** at MAIS Hacks 2023 Oct. 2023
- Won first place among 300+ hackers by creating an apartment-hunting web app using **machine learning**, **web scraping**, and **data analysis**
 - Intelligently filtered a rental API using an AI **image captioning** model to analyze listing photos and highlight top apartments

SKILLS

Languages: Python, Java, C, C++, JavaScript, Kotlin, Rust, BASH
Frameworks/Libraries: Flask, FastAPI, PyTorch, Tensorflow, React
Tools: Git, Docker, Linux